The Calabi invariant of a Hamiltonian isotopy is twice the total Hamiltonian, under the normalisation

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Editorial note. The counted statement is the author's labelled lemma 'l:Calabi' (source lines 485-491); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's normalisation condition on the Hamiltonian, whose label is invoked at the end of the retained proof.

Editorial note. The retained preamble is a minimal editorial adapter supplying only the plain text abbreviations that the publisher class used to define, followed by the native author macro block of the pinned source.

Editorial note. Every label used inside the retained text is defined inside one of the retained spans, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the Calabi-invariant identity for normalised Hamiltonian isotopies (author label l:Calabi).

Allowed standard prerequisites (editorial).

- Normalised Hamiltonian. A Hamiltonian H_t generating the isotopy is normalised when the integral of H_t over the boundary vanishes for every t.. It is the condition that kills the boundary term at source lines 506-508.
- The one-form a_{phi,nu}. a_{phi,nu} is the action one-form of the isotopy, equal to the integral over t of (X_t interior nu + H_t) composed with phi_t.. It is the integrand rewritten at source lines 494-500.
- Stokes theorem on a surface with boundary. The integral of an exact two-form over the surface equals the integral of the primitive over the boundary.. It is applied at source lines 501-505 to rewrite the integral of nu wedge dH_t.

$$\int_0^1 H_t(z)\,dt = 0, \qquad \forall z \in \partial\Sigma. \tag{1}$$

Lemma 0.1. *For each $\tilde{\phi} = [(\phi_t)] \in \widetilde{\mathrm{Ham}}_0(\Sigma,\omega)$, if H_t is the Hamiltonian normalized by* (1) *generating the isotopy ϕ_t, we have*

$$\int_\Sigma a_{\tilde{\phi},\nu}\,\omega = 2\int_0^1 \left(\int_\Sigma H_t\,\omega\right) dt.$$

Proof. From the identity $a_{\tilde{\phi},\nu} = a_{H,\nu}$ we find

$$\begin{aligned}\int_\Sigma a_{\tilde{\phi},\nu}\,\omega &= \int_\Sigma \left(\int_0^1 \big(X_t \lrcorner \nu + H_t\big) \circ \phi_t \, dt\right) \omega = \int_0^1 \left(\int_\Sigma \big(X_t \lrcorner \nu + H_t\big) \circ \phi_t \, \omega\right) dt \\ &= \int_0^1 \left(\int_\Sigma \big(X_t \lrcorner \nu + H_t\big)\, \omega\right) dt = \int_0^1 \left(\int_\Sigma \nu \wedge dH_t + H_t\, \omega\right) dt,\end{aligned}$$

where we have used the fact that ϕ_t preserves ω, and the identity $(X_t \lrcorner \nu)\omega = \nu \wedge dH_t$. By Stokes theorem, we find

$$\begin{aligned}\int_0^1 \left(\int_\Sigma \nu \wedge dH_t\right) dt &= \int_0^1 \left(\int_\Sigma \big(H_t\, d\nu - d(H_t \nu)\big)\right) dt \\ &= \int_0^1 \left(\int_\Sigma H_t\, \omega\right) dt - \int_0^1 \left(\int_{\partial\Sigma} H_t \nu\right) dt,\end{aligned}$$

and the latter integral vanishes thanks to the normalization condition (1):

$$\int_0^1 \left(\int_{\partial\Sigma} H_t \nu\right) dt = \sum_{C \in \pi_0(\partial\Sigma)} \left(\int_0^1 H_t(C)\, dt\right) \left(\int_C \nu\right) = 0. \qquad \square$$

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